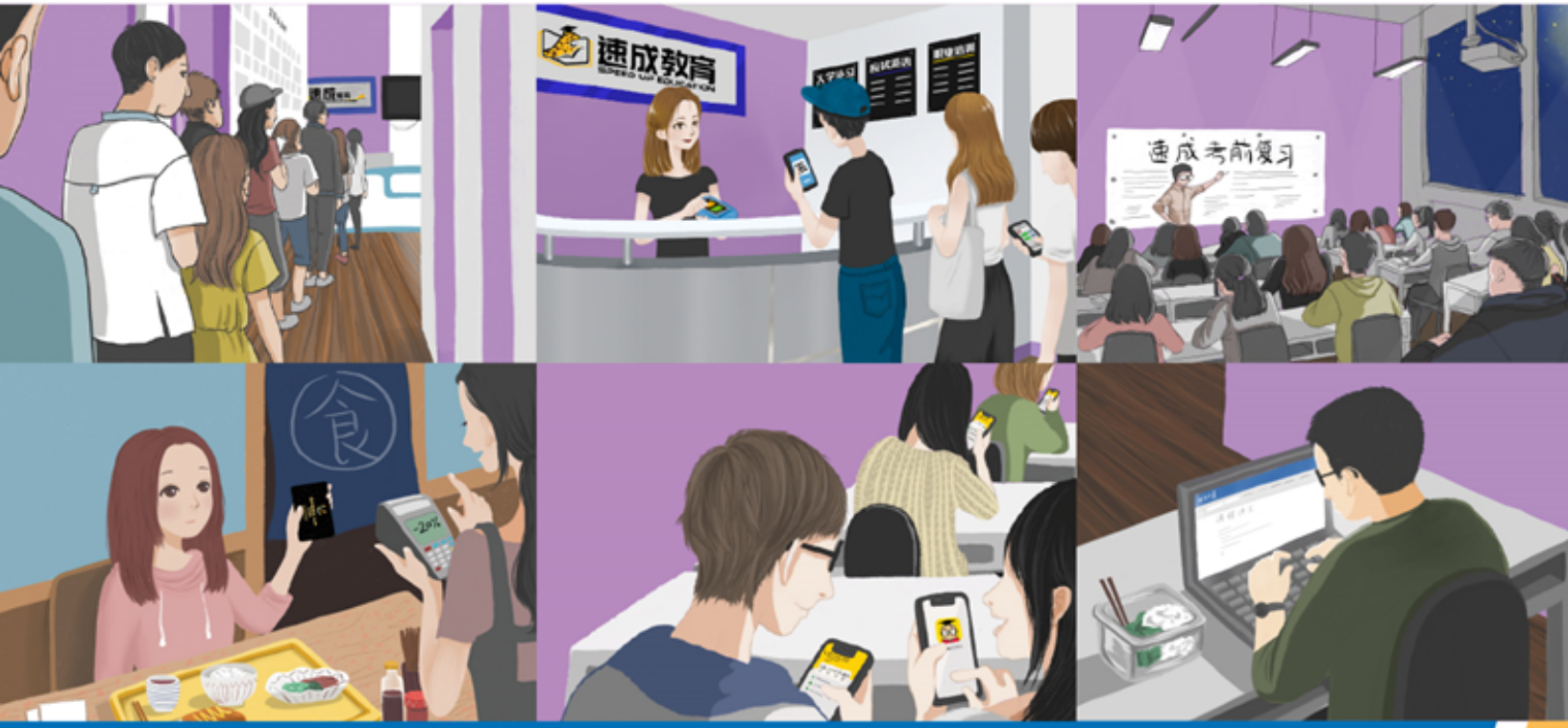
速成教育
SPEED UP EDUCATION



APSC 160

Introduction to Computation in Engineering Design

导师: **VC**

UBC Week 02 2026/9/18

Boolean

0 是 false，以外的数字都是 true，但一般用 1 表示 true

A	B	A && B	A || B	!A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

```
printf("%d\n", 1 > 2);                  -> 0
printf("%d\n", 2 <= 3);                 -> 1
printf("%d\n", 3 == 3.0);               -> 1
printf("%d\n", 3 != 3.0);               -> 0
int a = 1;

printf("%d\n", !a);                     -> 0

#include <stdbool.h>
bool b = true;
printf("%d\n", b);                      -> 1
printf("%d\n", 2 > 1);                  -> 1
printf("%d\n", b || 3 / 0 == 0);        -> 1
printf("%d\n", false && 1 + "2" == 3);  -> 0
printf("%d\n", true + true + false);    -> 2

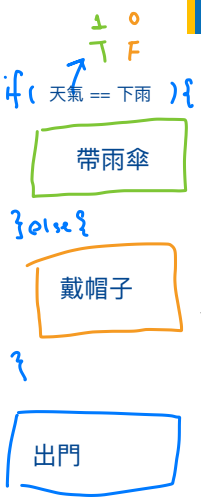
int is_gay;
printf("Are you gay? (1 - Yes | 0 - No)");
scanf("%d", &is_gay);
```

用 scanf() 還是只能用 int 去表達 boolean

考试必考

```
printf("%d\n", 3 - 2 || 3 / 0 == 0);     ->
printf("%d\n", 3 - 2 && 3 / 0 == 0);     ->
printf("%d\n", !(0.0 && 2 > 1) || 3 / 0 == 0);     ->
```

If Statement




```
int num = 12;
if (num > 10) {
    printf("Large");
} else {
    printf("Small");
}
print("Done");
```

Large
Done

```
int num = 8;
if (num > 10) {
    printf("Large");
} else {
    printf("Small");
}
printf("Done");
```

Small
Done

```
int num = 12
if (num > 10) {
    print("Large");
}

if (num >= 5) {
    print("Medium");
} else {
    print("Small");
}
print("Done");
```

Large
Medium
Done

```
int num = 12;
if (num > 10) {
    printf("Large")
} else if (num >= 5) {
    printf("Medium");
} else {
    printf("Small");
}
printf("Done");
```

Large
Done

```
int num = 8
if (num > 10) {
    print("Large");
} else if (num >=5) {
    print("Medium");
} else (
    print("Small");
}
print("Done");
```

Medium
Done

```
int num = 12
if (num >= 5) {
    print("Medium")
} else if (num > 10) {
    print("Large")
} else {
    print("Small");
}
print("Done");
```

Medium
Done

Nested If statements:

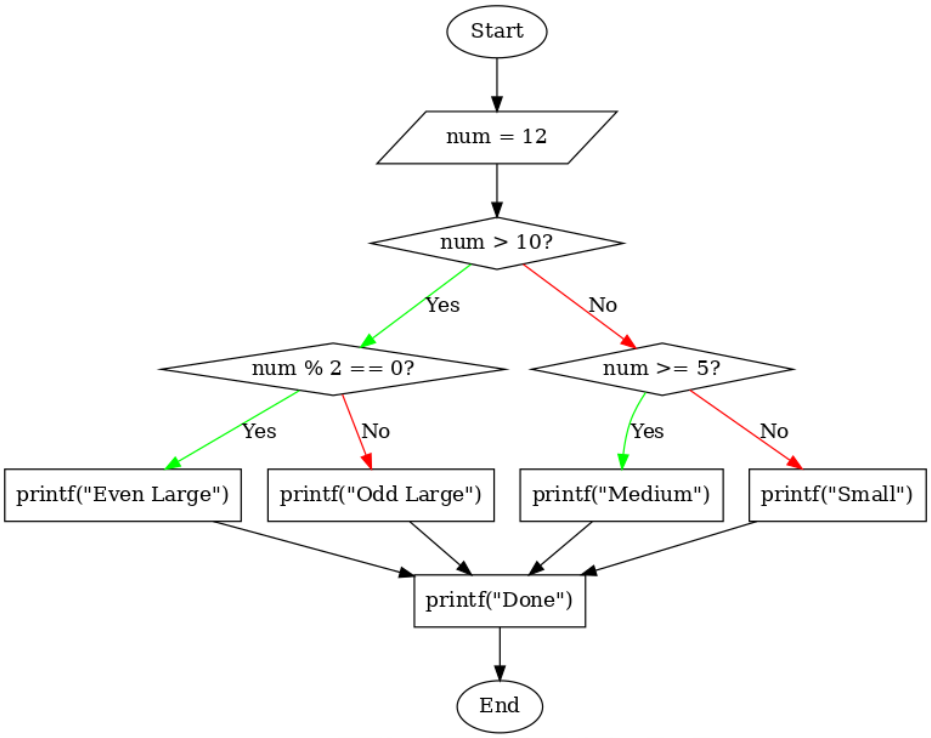
```
int num = 12;
if (num > 10) {
    if (num % 2 == 0) {
        printf("Even Large\n");
    } else {
        printf("Odd Large\n");
    }
} else if (num >= 5) {
    printf("Medium\n");
} else {
    printf("Small\n");
}
printf("Done\n");
```

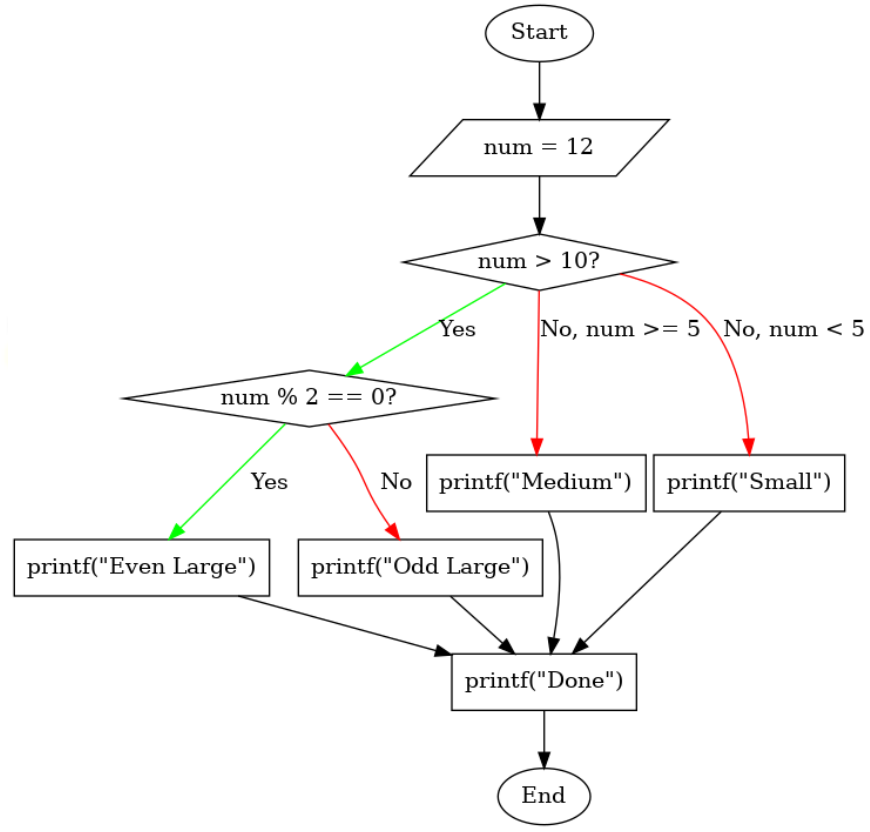
Simplify

```
if (x > 1) {
    if (y > 2) {
        if (z > 3) {
            printf("Hello\n");
        }
    }
}
```

```
if (x > 1 &&  y > 2 &&  z > 3) {
    print("Hello\n");
}
```

Control Flow Nested If statements:





If Statement Practices

考试练习题目 1

Output the practice time (in minute) for a player of the given age who may or may not be a beginner player, according to the practice times in the following table:

age of player	practice time
under 6 years	30 minutes
6 to 8 years, inclusive	50 minutes
over 8 years	75 minutes

Add 15 minutes to the practice time for a player who is a **not** beginner

Examples:

```
> Enter your age: 8
> Enter 1 if you are a beginner, 0 otherwise: 0
65
> Enter your age: 9
> Enter 1 if you are a beginner, 0 otherwise: 1
75
```

```c
#include <stdio.h>
#include <stdbool.h>

int main(void) {
    int age, beginner;

    printf("Enter your age: ");
    scanf("%d", &age);

    printf("Enter 1 if you are a beginner, 0 otherwise: ");
    scanf("%d", &beginner);

    int time;

    if (age < 6) {
       time = 30;
    } else if (age >= 6 && age <= 8) {
       time = 50;
    } else {
       time = 75;
    }

    if (!beginner == false) {
       time = time + 15;
    }

    printf("You need to practice %d minutes\n", time);
```

```
+=
-=
*=
/=
```

```
time += 15;
```

考试练习题目 2

The World Health Organization considers a person with Body Mass Index (BMI) less than 16 to be *severely underweight*. A non-severely underweight person with BMI less than 18.5 is considered *underweight*. A person with BMI of at least 30 is considered *obese*. A non-obese person with BMI of at least 25 is categorized as *overweight*, and everyone else is considered *normal*.



constant

```c
#include <stdio.h>
#include <stdbool.h>

#define OBESE "obese"
#define OVERWEIGHT "overweight"
#define NORMAL "normal"
#define UNDERWEIGHT "underweight"
#define SEVERELY_UNDERWEIGHT "severely underweight"

int main(void) {
    double bmi;

    printf("Enter your bmi, you FAT PIG: ");
    scanf("%lf", &bmi);

    if (bmi <= 16.0) {
       printf(SEVERELY_UNDERWIEGHT);
    } else if (...) {
       ...
    }
```

考试练习题目 3

Output the year of study for a student who has taken `num_of_courses` courses

- Students with 10 courses or under are in first-year
- Students who have taken between 11 to 20 courses are in second-year
- Students who have taken 21 to 30 courses are in third-year
- Students who have taken 31 to 40 courses are in fourth-year
- Students who have taken more than 40 courses are considered fifth-year or more

```
#include <stdio.h>
#include <stdbool.h>

int main(void) {
    int num_of_courses;

    printf("Enter the number of courses you've taken: ");
    scanf("%d", &num_of_courses);

    if (                                                    ) {
       if (                                                 ) {
          printf("fifth-year or more\n");
       } else {
          if (                                              ) {
             printf("third-year\n");
          } elif (                                          ) {
             printf("fourth-year\n");
          }
          printf("second-year\n");
       }
    } else {
       printf("first-year\n");
    }

    return 0;
}
```

用数学的方式表达以下代码当 x 为多少的时候结果为 2

```
if (x < 20) {
    printf("1");
} else if (x > 15 && x < 30) {
    printf("2");
} else {
    printf("3");
}
```

[20, 30)

更多考试练习题目（用 VSCode）

题目 1：票价计算器

一个电影院的票价根据以下规则计算：

- 年龄 < 6 → 免费
- 年龄 6–18 → 8 元
- 年龄 19–64 → 12 元
- 年龄 ≥ 65 → 6 元
- 如果 是学生，则票价再打 8 折。
- 如果 不是学生，且 周末，票价再加 2 元。

请写程式输入年龄、是否为学生（1/0）、是否周末（1/0），输出票价。

题目 2：税率计算

某国税制如下：

- 收入 ≤ 10,000 → 不课税
- 收入 10,001–40,000 → 10% 税率
- 收入 40,001–80,000 → 20% 税率
- 收入 > 80,000 → 30% 税率
- 如果 是老年人 → 减免 2,000 税额。
- 如果 是残障人士 → 税率再减 5%。

输入收入、是否老年人（1/0）、是否残障人士（1/0），输出应缴税额。

题目 3：运动员资格判断 🏃

某比赛规则如下：

- 年龄必须 ≥ 18
- BMI < 30 才能参加
- 如果 有医生证明 (boolean)，则可以忽略 BMI 限制。
- 如果 是专业选手 (boolean)，年龄限制放宽到 16 岁。

输入年龄、BMI、是否有医生证明 (1/0)、是否专业选手 (1/0)，输出是否符合资格。

题目 4：游乐园票价 🎢

- 基本票价 = 100 元
- 身高 < 120 cm → 半价
- 身高 > 200 cm → 禁止入场
- 如果 是 VIP → 免费
- 如果 不是 VIP 且 是假日 → 票价 +20%

输入身高 (cm)、是否 VIP (1/0)、是否假日 (1/0)，输出票价或禁止讯息。

题目 5：贷款审核系统 🏦

规则如下：

- 收入 < 20,000 → 拒绝
- 收入 ≥ 20,000 → 进入进一步审查：
  - 如果 信用分数 ≥ 700 → 批准
  - 如果 信用分数 600–699 → 若 有担保人 (boolean) → 批准，否则拒绝
  - 如果 信用分数 < 600 → 拒绝
- 如果 收入 ≥ 100,000 且信用分数 ≥ 750 → 自动批准（不需要担保人）。

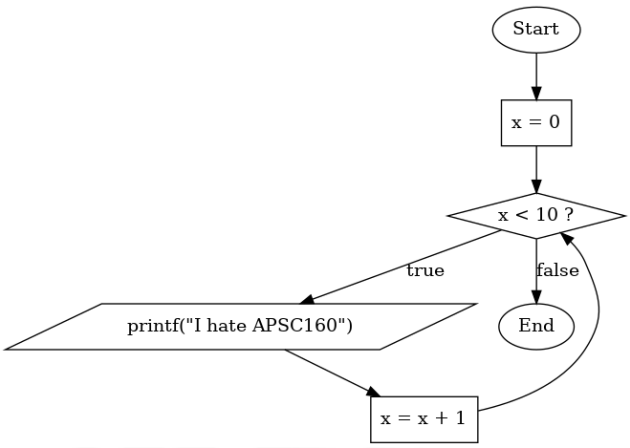
输入收入、信用分数、是否有担保人 (1/0)，输出「批准」或「拒绝」。

Loops

Assume we want to print 'I hate APSC160' ten times, we can repeat the code ten times:

```
printf("I hate APSC160");
printf("I hate APSC160");
printf("I hate APSC160");
...
...
printf("I hate APSC160");
```

10 次



A better solution will be using **while loop**

```
int x = 0;
while (x < 10) {
    printf("I hate APSC160");
    x = x + 1;
}
```

You can also change it to a **for loop**

进入循环之前 跑一次　　每次循环之前 检查

```
for(int x = 0; x < 10; x = x + 1) {
    printf("I hate APSC160");
}
```

每次循环结束 执行

```
for(int i=0; i<10; i=i+1) {
    printf("%d\n", i);
}
for(float j=10000; j>=.00001; j=j/10.0) {
    printf("%f\n", j);
}
for(char d='A'; d<'F'; d=d+1) {
    printf("%c , %d\n", d, d);
}
```

OutPut:

```
0
1
2
3
4
5
6
7
8
9
10000.000000
1000.000000
100.000000
10.000000
1.000000
0.100000
0.010000
0.001000
0.000100
A , 65
B , 66
C , 67
D , 68
E , 69
```

x = x + 1 可以写成 x += 1, 也可以写成 x++ 或 ++x;

那么 x++ 和 ++x 有何区别?

```
int a = 5;
printf("%d\n", x++);
printf("%d\n", x);
```

5 V.S. 6

```
int a = 5;
printf("%d\n", ++x);
printf("%d\n", x);
```

6 6

Short Cuts

```
var += ...
var -= ...
var *= ...
var /= ...
```

Example

```
/* Print all odd numbers from 1 to n (inclusive). */
int main() {
    int n;
    scanf("%d", &n);
```

题型 1: 提前结束

```
/* Print ‘prime’ if val is prime number, otherwise print ‘not prime’. */
int main() {
    int val;
    scanf("%d", &val);
```

```
/* Print all prime numbers from 1 to 100. */
int main() {
```

题型 2: 累积 Accumulate

```
/* Print the number of all prime numbers from 1 to 100. */
int main() {
```

```c
/* Print the sum of all positive integers up to n (inclusive). */
int main() {
    int n;
    scanf("%d", &n);
```

```c
/* Print the sum of all integers between 1 and n that are divisible by 3 or 5.
 * Examples:
 *     if n = 10, then should print 33 (3+5+6+9+10)
 */
int main() {
    int n;
    scanf("%d", &n);
```

```
/* Return the number of even digits in num.
 * Examples:
 *     if num = 23641 then should print 3
 */
int main() {
    int num;
    scanf("%d", &num);
```

```
/* Print “Paldindrome” iff the given integer <num> reads the same forwards and
 * backwards. Print “Not Paldindrome” otherwise.
 * Examples:
 *     if num = 12321 then should print “Paldindrome”
 *     if num = 1231 then should print “Not Paldinrome”
 */
int main() {
    int num;
    scanf("%d", &num);
```